

One Pot Synthesis of γ -Benzopyranones via Iron-Catalyzed Aerobic Oxidation and Subsequent 4-Dimethylaminopyridine Catalyzed 6-*endo* Cyclization

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Abstract: One-pot synthesis of γ -benzopyranones was realized in decent yields and excellent regioselectivities via iron-catalyzed aerobic oxidation and 4-dimethylaminopyridine-catalyzed cyclization of related propargylic alcohols. Derivatizations to aromatic substituted γ -benzopyranones and synthesis of naturally occurring 3',4'-dimethoxyflavone have also been realized.

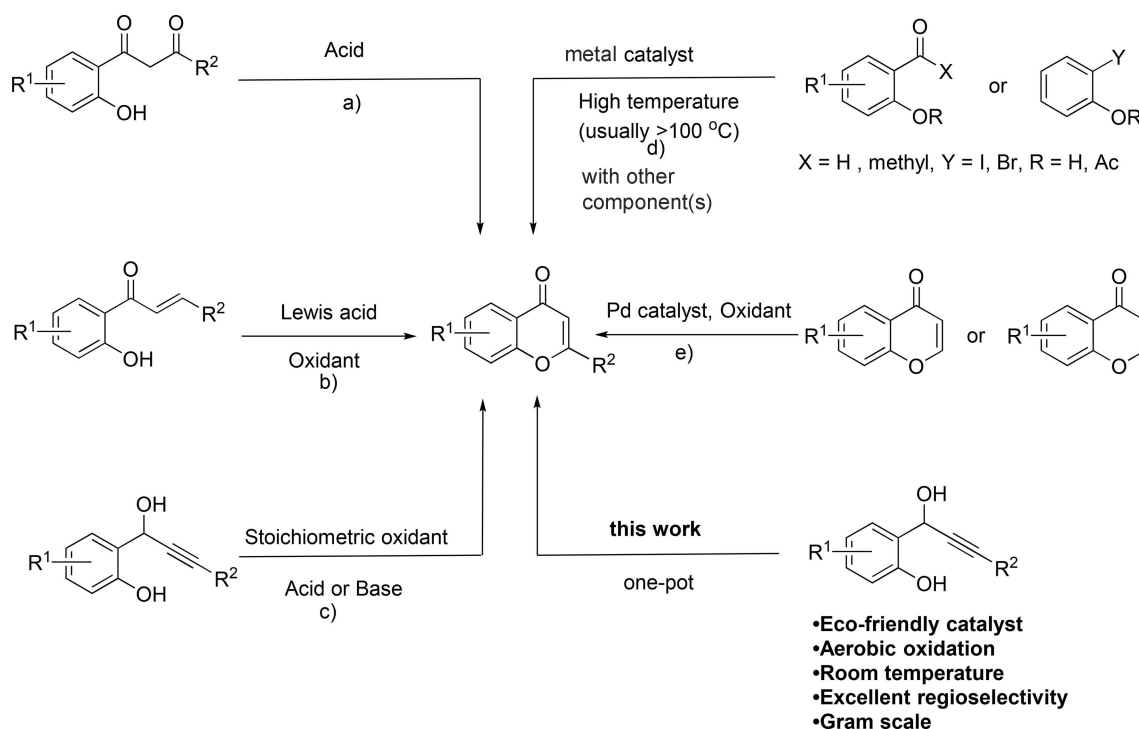
Keywords: one pot; iron catalysis; aerobic oxidation; flavones; natural products

As a major class of natural products commonly found in plants, γ -benzopyranones have received a lot of attention due to their biological activities such as anti-tumor and anti-inflammatory activities.^[1] In addition, the structure of conjugate enone fragment and fused phenyl have a great synthetic potential in organic synthesis.^[2] Thus, much attention has been focused on the methods for flavone synthesis. The reported literatures mainly consist of five types of strategies (Scheme 1): a) cyclization and dehydration of β -carbonyl compounds.^[3] b) Michael addition of chalcones and subsequent oxidation.^[4] c) Michael addition of alkynones.^[5] d) Metal-catalyzed cyclization reaction.^[6] e) Palladium-catalyzed oxidative coupling reaction.^[7] Unfortunately, they all suffer from too many steps, stoichiometric amount of oxidant, highly toxic carbon monoxide, expensive palladium catalyst or high temperature. The direct protocol using propargylic alcohols as starting material remains quite undeveloped.^[8] Recently, our group developed an

iron-catalyzed oxidation of different alcohols at room temperature in high yields and excellent selectivities.^[9] By using this method, the oxidation of propargylic alcohols can be realized in excellent yields and selectivities. We hypothesized that by adding an extra catalyst into the oxidation system, γ -benzopyranones can be generated from propargylic alcohols in a one-pot process. Herein, we report our results on this topic.

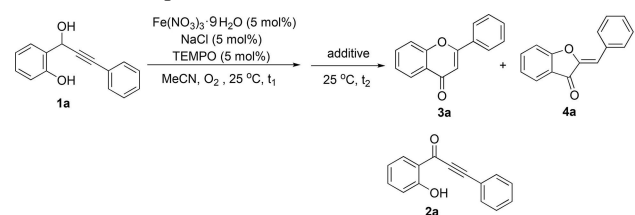
To start our study, 2-(1-hydroxy-3-phenylprop-2-yn-1-yl)phenol (**1a**) was selected as the model starting material. First, different additives were screened after the iron-catalyzed aerobic oxidation reaction.^[9] When K_2CO_3 and NaOH were used, both 2-phenyl-4H-chromen-4-one (**3a**) and 2-benzylidenbenzofuran-3(2H)-one (**4a**) were formed in moderate yields (Table 1, entries 1 and 2).^[10] Other additive such as PCy_3 (Table 1, entry 3) resulted in a poor selectivity and low conversion. TfOH and *t*-BuOLi gave 86% yield and 79% yield of alkynone **2a**, respectively (Table 1, entries 4 and 5). Running this reaction by adding DMAP^[11] gave the best result in 85% yield of **3a** with an excellent selectivity and no starting material was recovered (Table 1, entry 5).

Encouraged by these results, different solvents were examined. DMF, dioxane, and THF gave 43–73% recovery of **1a** (Table 2, entries 1–3). When DCM was used as the solvent, 8% yield of γ -benzopyranone **3a** was detected without recovery of **1a** (Table 2, entry 4). When DCE was used, 58% yield of **3a** was obtained (Table 2, entry 5). Finally, MeCN turned out to be the best solvent, providing γ -benzopyranone **3a** in 85% NMR yield and 82% isolated yield (Table 2, entry 6).



Scheme 1. The synthesis of γ -benzopyranones.

Table 1. The optimization of the reaction conditions.^[a]



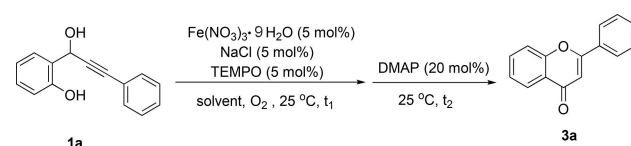
Entry	Additive	Equiv.	t_1/t_2 (h)	Yield (%) ^[b]		
				2a	3a	4a
1	K ₂ CO ₃	2.0	3.0/4.0	0	40	52
2	NaOH	2.0	3.0/2.5	0	15	61
3	PCy ₃	0.2	3.0/2.5	29	12	19
4	TfOH	0.2	2.3/2.5	86	0	0
5	<i>t</i> -BuOLi	0.2	3.2/14	79	0	0
6	DMAP	0.2	3.3/2.5	0	85	0

^[a] The reaction was conducted using **1a** (1.0 mmol), Fe(NO₃)₃·9H₂O (0.05 mmol), NaCl (0.05 mmol), TEMPO (0.05 mmol), and additive in MeCN (5 mL) with an oxygen balloon at 25 °C.

^[b] Determined by ¹H NMR analysis with dibromomethane as the internal standard.

Subsequently, different loadings of DMAP were studied in this one-pot synthesis. With 0.1 equiv. of DMAP, only 19% yield of **3a** was detected together with 67% yield of aerobic oxidation product 2-alkynone **2a** (Table 3, entry 1). When the loading of DMAP was increased to 0.15 equiv, the yield of **3a**

Table 2. The effect of solvent.^[a]



Entry	Solvent	t_1/t_2 (h)	Yield of 3a (%) ^[b]	Recovery of 1a (%) ^[b]
1	DMF	2.8/1.7	0	73
2	dioxane	2.7/2.4	0	43
3	THF	2.5/2.0	0	49
4	DCM	3.0/2.5	8	0
5	DCE	3.0/2.5	58	0
6	MeCN	3.3/2.5	85 (82)^[c]	0

^[a] The reaction was conducted using **1a** (1.0 mmol), Fe(NO₃)₃·9H₂O (0.05 mmol), NaCl (0.05 mmol), TEMPO (0.05 mmol) in solvent (5 mL) with an oxygen balloon at 25 °C. After the disappearance of **1a** by TLC, DMAP (0.2 mmol) was added into the resulting mixture.

^[b] Determined by ¹H NMR analysis with dibromomethane as the internal standard.

^[c] The number in parenthesis represents isolated yield. DMAP: 4-dimethylaminopyridine.

was improved to 72% with no **2a** detected (Table 3, entry 2). The best catalyst of DMAP was proved to be 0.2 equiv. with 85% NMR yield of **3a** and no **2a** was detected (Table 3, entry 4). Further increasing the

Table 3. The effect of the DMAP loading.^[a]

Entry	x	t ₁ /t ₂ (h)	Yield (%) ^[b]	
			2a	3a
1	10	3.25/2.5	67	19
2	15	3.0/2.5	0	72
3	20	3.3/2.5	0	85 (81)^[c]
4	30	2.6/1.5	0	83

^[a] The reaction was conducted using **1a** (1.0 mmol), Fe(NO₃)₃·9H₂O (0.05 mmol), NaCl (0.05 mmol), TEMPO (0.05 mmol) in MeCN (5 mL) with an oxygen balloon at 25 °C. After the disappearance of **1a** by TLC, DMAP (0.2 mmol) was added into the resulting mixture.

^[b] Determined by ¹H NMR analysis with dibromomethane as the internal standard.

^[c] The number in parenthesis represents isolated yield.

loading of DMAP failed to improve the yield of **3a** (Table 3, entry 5).

With the optimal conditions in hand (Table 3, entry 3), the reaction scope was further investigated. When R¹ is H, different aryl groups can be introduced into the R² position including electron withdrawing and electron donating group substituted phenyl groups, providing the desired γ-benzopyranones in decent yields (Table 4, entries 1–4, **3a–3d**). Replacing R² with H and *n*-Bu resulted in lower yields (Table 4, entries 5 and 6, **3e** and **3f**). To understand the reason of low yields for the substrates **1e** and **1f**, iron-catalyzed oxidation and 6-*endo* cyclization were conducted separately. Firstly, the oxidation of **1e** provided the desired product **2e** in 83% NMR yield. The second step of 6-*endo* cyclization gave only 33% NMR yield of **3e**. Thus, we reasoned that the low yield of **3e** was mainly due to the low yield of 6-*endo* cyclization step. For the reaction of **1f**, iron-catalyzed oxidation provided 89% NMR yield of **2f**, the second step of 6-*endo* cyclization provided 86% NMR yield of **3f** with acetonitrile as solvent.^[11] However, the 6-*endo* cyclization under the catalysis of DMAP together with the catalysts for the first step gave **3f** in only 53% NMR yield. Based on these results, we proposed that the low yield of **3f** was also caused by the low yield of 6-*endo* cyclization under the combined conditions (Scheme 2).

R¹ may also be 5-chloro with R² being phenyl, cyclopropyl, 2-thienyl, 3-thienyl, providing the desired product in 52–86% isolated yields (Table 4, entries 7–10, **3g–3j**). It is worth noting that methyl, bromo, nitro, and 3,5-dichloro on R¹ position were all tolerated in this reaction providing the corresponding products in good to excellent yields (Table 4, entries

Table 4. The substrate scope.^[a]

Entry	R ¹ /R ²	t ₁ /t ₂ (h)	Yield of 3 (%)
1	H/Ph (1a)	3.3/2.5	82 (3a)
2	H/ <i>p</i> -MeC ₆ H ₄ (1b)	3.0/3.0	81 (3b)
3 ^[b]	H/ <i>p</i> -MeOC ₆ H ₄ (1c)	3.75/5.5	81 (3c)
4	H/ <i>p</i> -ClC ₆ H ₄ (1d)	3.3/11.6	68 (3d)
5	H/H (1e)	4.0/3.0	20 (3e)
6	H/ <i>n</i> -Bu (1f)	3.0/3.0	44 (3f)
7	5-Cl/Ph (1g)	3.5/3.0	78 (3g)
8	5-Cl/Cyclopropyl (1h)	3.5/3.5	86 (3h)
9	5-Cl/2-thienyl (1i)	3.5/3.5	52 (3i)
10	5-Cl/3-thienyl (1j)	3.5/4.0	78 (3j)
11	5-Me/Ph (1k)	3.5/3.0	67 (3k)
12	5-Br/Ph (1l)	3.75/13	78 (3l)
13	5-NO ₂ /Ph (1m)	3.75/13	85 (3m)
14	3-Cl, 5-Cl/Ph (1n)	12/7.0	63 (3n)
15	5-MeO/Ph (1o)	13/4.0	13 (3o)

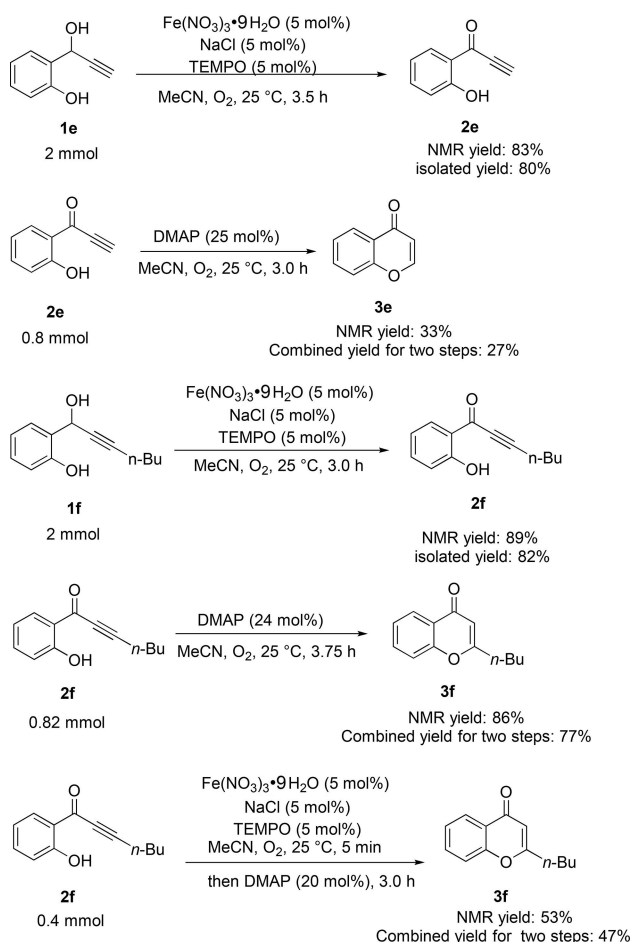
^[a] The reaction was conducted using **1a** (1.0 mmol), Fe(NO₃)₃·9H₂O (0.05 mmol), NaCl (0.05 mmol), TEMPO (0.05 mmol) in MeCN (5 mL) with an oxygen balloon at 25 °C. After the disappearance of **1a** by TLC, DMAP (0.2 mmol) was added into the resulting mixture.

^[b] The reaction was conducted using **1c** (0.5 mmol).

11–14, **3k–3n**). However, when R¹ is 5-methoxy group, the reaction provided the desired product **3o** in only 13% isolated yield (Table 4, entry 15, **3o**).

Since γ-benzopyranones with multi-methoxy substituents are core structure of natural products and play an important role in pharmaceutical science, we also tested 4-methoxy and 4,5-dimethoxy substituted propargylic alcohols (**1p** and **1q**). Since **1p** and **1q** are easily decomposed during the chromatographic purification, we used the crude products directly for the reaction. However, the one-pot process of oxidation and cyclization for **1p** provided only 10% yield of **3p**, and one-pot synthesis of **3q** failed, yielding a complex mixture-product **3q** was not observed from the crude NMR spectra. The poor results were due to the oxidation steps (24% NMR yield for **2p** and 5% NMR yield for **2q**). When **2q** was isolated and submitted for the cyclization process, the reaction proceeded smoothly to provide the final product **3q** in 76% yield (Scheme 3).

To demonstrate the practicality and efficiency of this methodology, gram scale synthesis of γ-benzopyranones were carried out. When **1l** and **1n** were chosen to be the substrates, 1.53 g of **3l** (85% yield) and 1.07 g of **3n** (61% yield) were successfully prepared (Scheme 4).



Scheme 2. Reactions of **1e/2e** and **1f/2f**.

Naturally occurring 3',4'- dimethoxyflavone **3r** was chosen as the target and synthesized in four steps from commercially available starting materials (Scheme 4).^[12] The first two steps including Sonogashira coupling and subsequent Retro-Favorski reaction gave **6r** in 55% yield. Then, the reaction of salicylic aldehyde with **6r**-based lithium acetylide gave the desired propargylic alcohol **1r** in 85% yield. Finally, the naturally occurring 3',4'-dimethoxyflavone **3r** was obtained in 78% yield (Scheme 5).

The Suzuki coupling reaction of **3n** with phenylboronic acid gave the desired γ -benzopyranone **7n** by using LB-Phos^[13] as the ligand in 71% isolated yield via simple recrystallization (Scheme 6).

In conclusion, a green methodology consisting of iron-catalyzed aerobic oxidation and DMAP-catalyzed 6-*endo* cyclization to differently substituted γ -benzopyranones in moderate to excellent yields was developed. This eco-friendly one-pot synthesis protocol may be easily executed on gram scale and has been applied to the synthesis of naturally occurring 3',4'-dimethoxyflavone **3r**. Due to the importance of these products, this method will be of high interest to

people who work in fine chemical and pharmaceutical synthesis.

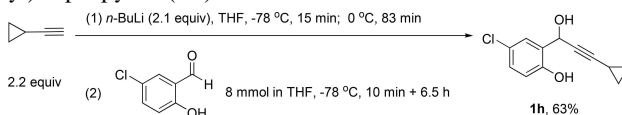
Experimental Section

General Information. NMR spectra were taken with an Agilent-400 spectrometer (400 MHz for ¹H NMR, 100 MHz for ¹³C NMR). ¹H NMR spectra were recorded using TMS as an internal standard (δ 0 ppm), or using solvent residual signal of acetone-d₆ as an internal standard (δ 2.05 ppm), or using solvent residual signal of CD₃CN as an internal standard (δ 1.95 ppm) and coupling constants were reported in Hz. ¹³C NMR spectra were recorded using CDCl₃ as an internal standard (δ 77.0 ppm), or using acetone-d₆ as an internal standard (δ 29.84 ppm), or using CD₃CN as an internal standard (δ 118.2 ppm). All reactions were carried out in flame-dried Schlenk tubes. Fe(NO₃)₃·9H₂O (98%) and Pd(OAc)₂ (47.5% Pd) was purchased from J&K Chemicals; NaCl (>99.5%), CuI (>99.5%) and DMAP (>99%) was purchased from Sinopharm Chemical Reagent Co., Ltd; TEMPO (98%) was purchased from Shanghai Darui Fine Chemical Co., Ltd.; Pd(PPh₃)₂Cl₂ (98%) was purchased from Energy Chemical; Dioxane, toluene and THF were dried over sodium wire. MeCN was dried over CaH₂ and distilled. All the temperatures are referred to the oil baths, acetone/dry ice bath or ice/water bath used. Recovery of substrates was determined by ¹H NMR analysis using dibromomethane as the internal standard.

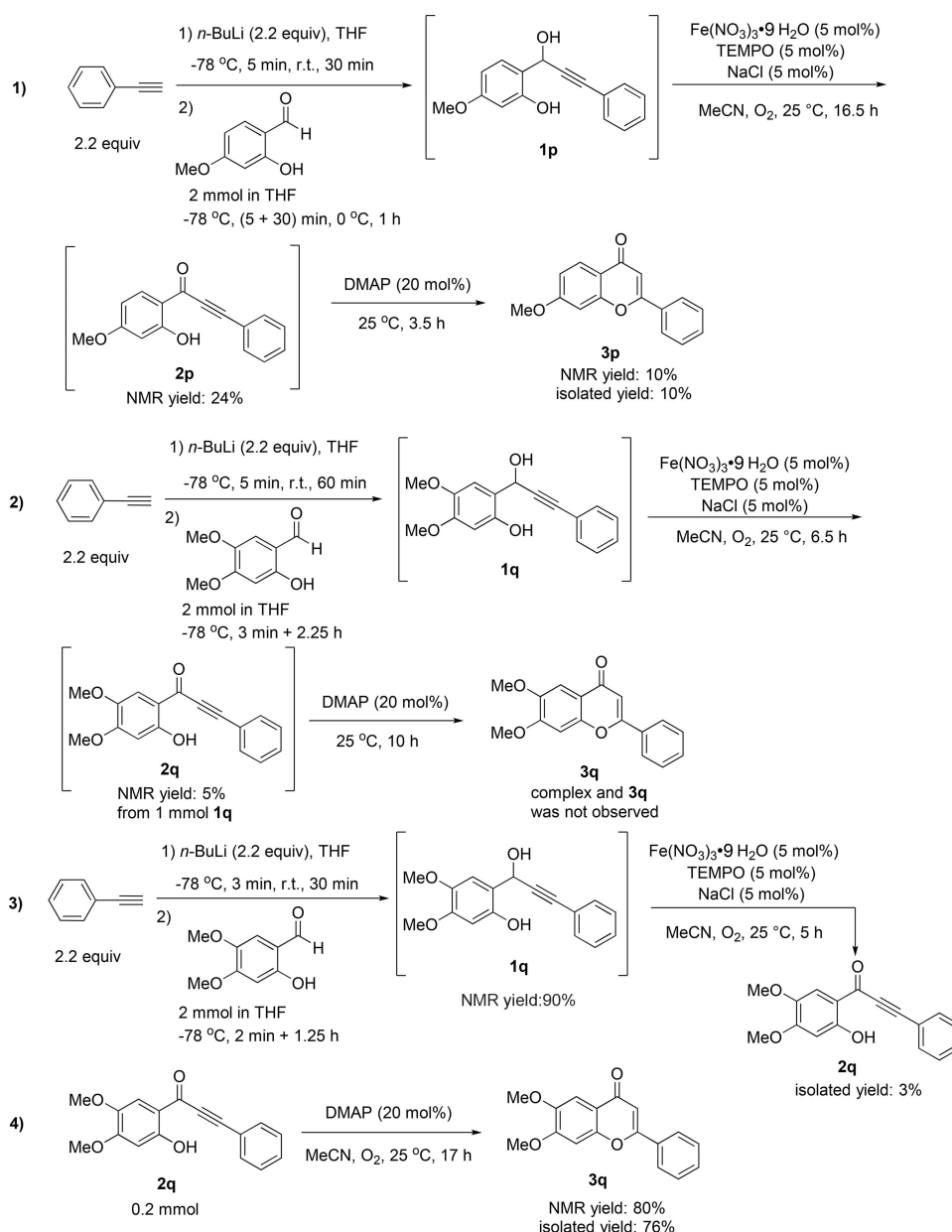
Experimental Details and Analytical Data

1. Synthesis of Propargylic Alcohols^[8]

(1) Preparation of 3-cyclopropyl-1-(2-hydroxy-5-chlorophenyl)-2-propynol (**1h**)



Typical Procedure A: To a flame-dried Schlenk tube were added cyclopropyl acetylene (1.18 g, 17.6 mmol) and THF (40 mL) under Ar. The resulting mixture was cooled to -78°C . *n*-BuLi (2.4 M in hexanes, 7 mL, 16.8 mmol) was slowly added at -78°C within 15 min. The mixture was put into an ice-water bath and stirred at 0°C for 83 min. Then, the mixture was put into a dry ice-acetone bath at -78°C . After 10 min, the solution of 5-chloro-2-hydroxybenzaldehyde (1.20 g, 8 mmol) in dry THF (20 mL) was slowly added within 10 min. After being stirred at -78°C for 6.5 h, the resulting mixture was placed in an ice/water bath and quenched with a saturated aqueous ammonium chloride solution (40 mL). The organic layer was separated and the aqueous solution was extracted with ethyl ether (3×30 mL). The combined organic layer was dried over anhydrous Na₂SO₄. After filtration and concentration under reduced pressure, the crude product was purified by column chromatography on silica gel to afford **1h** (1.0735 g, 63%) as a white solid [DCM (100 mL) to DCM/ethyl ether=15/1 afforded impure **3h**, which was further purified with chromatography (eluent: petroleum ether/ethyl acetate=3/1)]; m.p. 74.0 –

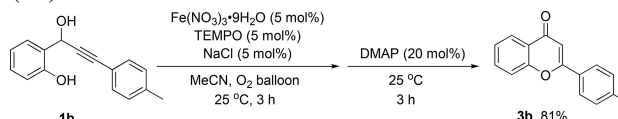


Scheme 3. Reactions of multi-methoxy substituted substrates **1p** and **1q**.

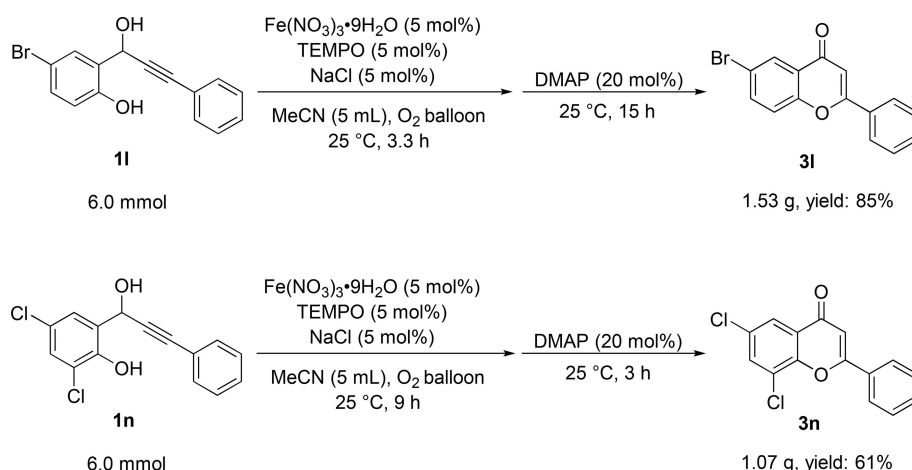
74.9 °C (petroleum ether/dichloromethane); ^1H NMR (400 MHz, CDCl_3): δ = 7.36 (s, 1 H, ArOH), 7.31 (d, J = 2.4 Hz, 1H, ArH), 7.18 (dd, J_1 = 8.4 Hz, J_2 = 2.4 Hz, 1H, ArH), 6.83 (d, J = 8.8 Hz, 1H, ArH), 5.59 (d, J = 4.0 Hz, 1H, OCH), 2.68 (d, J = 5.2 Hz, 1H, OH), 1.40–1.30 (m, 1H, CH from cyclopropane), 0.90–0.72 (m, 4H, $\text{CH}_2\text{-CH}_2$); ^{13}C NMR (100 MHz, CDCl_3): δ = 153.9, 129.7, 127.4, 126.2, 124.8, 118.4, 93.3, 72.3, 63.8, 8.5, -0.5 ; MS (70 eV, EI) m/z (%): 224 (M^+ (^{37}Cl), 2.55), 222 (M^+ (^{35}Cl), 8.51), 176 (100); IR (neat): ν = 3394, 2232, 1602, 1494, 1424, 1358, 1326, 1271, 1113 cm^{-1} ; Anal. Calcd for $\text{C}_{12}\text{H}_{11}\text{ClO}_2$: C 64.73, H 4.98; Found: C 64.84, H 4.98.

2. Synthesis of γ -Benzopyranones

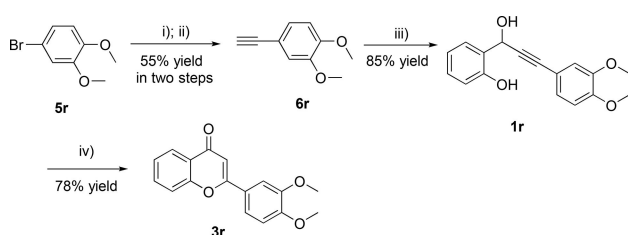
(1) Preparation of 2-(4-methylphenyl)-4H-chromen-4-one (**3b**)



Typical Procedure B: To a flame-dried Schlenk tube were added $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (20.1 mg, 0.05 mmol) and NaCl (3.1 mg, 0.05 mmol) under argon. The Schlenk tube was then degassed to remove the argon inside completely, and refilled with O_2 by a balloon of O_2 for three times. Then, a solution of TEMPO (8.2 mg, 0.05 mmol) in dry MeCN (3 mL) and a

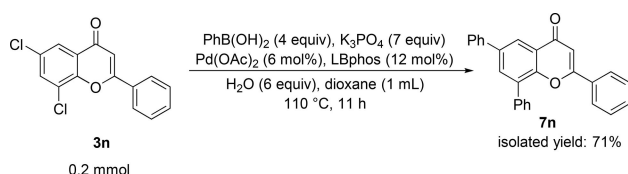


Scheme 4. Gram scale reactions.



Scheme 5. Synthesis of naturally occurring 3',4'-dimethoxyflavone **3r**.

Reagents and conditions: i) 2-methylbut-3-yn-2-ol (3.6 equiv.), Pd(PPh₃)₂Cl₂ (5 mol%), CuI (10 mol%), TEA, 90 °C, 24 h; ii) KOH (3 equiv.), toluene, 110 °C, 22 h; iii) salicylic aldehyde (0.45 equiv.), *n*-BuLi (0.95 equiv.), THF, −78 °C, 4 h; iv) Fe(NO₃)₃·9H₂O (5 mol%), NaCl (5 mol%), TEMPO (5 mol%), MeCN, oxygen balloon, 25 °C, 4.7 h; i) then DMAP (20 mol%), 25 °C, 6 h.

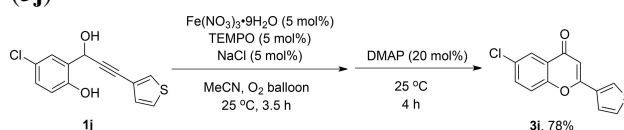


Scheme 6. Synthetic application.

solution of **1b** (239.4 mg, 1.0 mmol) in dry MeCN (2 mL) were added sequentially. Then the mixture was stirred at 25 °C with a balloon of O₂ for 3.0 h. After the disappearance of **1b** by TLC, O₂ balloon was removed and DMAP (24.1 mg, 0.2 mmol) was added into the reaction mixture. Then the mixture was stirred at 25 °C for 3.0 h as monitored by TLC. After concentration under reduced pressure, the residue was diluted with DCM. To the solution was added 1 M HCl (10 mL) and the resulting mixture was washed with 20 mL of water. The organic layer was separated and the aqueous solution was extracted with DCM (3 × 30 mL). The combined organic layer was dried over anhydrous Na₂SO₄,

filtered and concentrated under reduced pressure. Then, the mixture was transferred with DCM, filtered through a short column on silica gel (3 cm) (eluent: ethyl ether) until no product can be observed by TLC, and concentrated under reduced pressure [the yield of **3b** was determined by ¹H NMR analysis of the crude products using dibromomethane as the internal standard (82%)]. The residue was purified by column chromatography on silica gel to afford **3b**^[8] (193.5 mg, 81%) as a pale yellow solid [eluent: petroleum ether/ethyl acetate = 8/1]; m.p. 112.5–113.4 °C (petroleum ether/DCM) (Lit^[6f] m.p.: 111 °C); ¹H NMR (400 MHz, CDCl₃): δ = 8.21 (dd, *J*₁ = 7.6 Hz, *J*₂ = 1.2 Hz, 1H, ArH), 7.78 (d, *J* = 8.4 Hz, 2H, ArH), 7.70–7.62 (m, 1H, Ar–H), 7.53 (d, *J* = 8.4 Hz, 1H, ArH), 7.38 (t, *J* = 7.6 Hz, 1H, ArH), 7.28 (d, *J* = 7.6 Hz, 2H, ArH), 6.76 (s, 1H, =CH), 2.40 (s, 3 H, CH₃); ¹³C NMR (100 MHz, CDCl₃): δ = 178.3, 163.4, 156.1, 142.1, 133.5, 129.6, 128.8, 126.1, 125.5, 125.0, 123.8, 117.9, 106.8, 21.4; MS (70 eV, EI) *m/z* (%): 236 (M⁺, 100); IR (neat): ν = 1634, 1604, 1567, 1510, 1464, 1414, 1370, 1124 cm^{−1}.

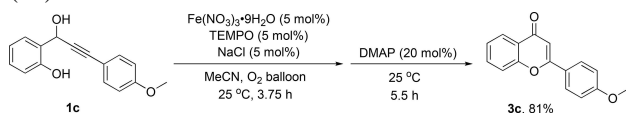
(2) Preparation of 6-chloro-2-(thien-3-yl)-4H-chromen-4-one (**3j**)



Typical Procedure C: To a flame-dried Schlenk tube were added Fe(NO₃)₃·9H₂O (20.2 mg, 0.05 mmol) and NaCl (2.7 mg, 0.05 mmol) under argon. The Schlenk tube was then degassed to remove the argon inside completely, and refilled with O₂ by a balloon of O₂ for three times. Then, a solution of TEMPO (8.1 mg, 0.05 mmol) in dry MeCN (2 mL) and a solution of **1j** (264.7 mg, 1.0 mmol) in dry MeCN (3 mL) were added sequentially. Then the resulting mixture was stirred at 25 °C with a balloon of O₂ for 3.5 h. After the disappearance of **1j** by TLC, O₂ balloon was removed and DMAP (24.4 mg, 0.2 mmol) was added into the reaction mixture. Then the mixture was stirred at 25 °C for 4 h as monitored by TLC. After concentration under reduced pressure, the residue was diluted with DCM. To the solution

was added 1 M HCl (10 mL) and the resulting mixture was washed with 20 mL of water. The organic layer was separated and the aqueous solution was extracted with DCM (3 × 30 mL). The combined organic layer was dried over anhydrous Na₂SO₄, filtered and concentrated to remove most of DCM. Then, the resulting mixture was filtered through a short column on silica gel (3 cm) (eluent: ethyl ether) until no product may be collected as indicated by TLC, and concentrated under reduced pressure [the yield of **3j** was determined by ¹H NMR analysis of the crude products using dibromomethane as the internal standard (83%)]. The residue was purified by column chromatography on silica gel to afford **3j**^[14] (207.2 mg, 78%) as a white solid [eluent: petroleum ether/ethyl acetate = 15/1 (400 mL) to petroleum ether/dichloromethane/ethyl acetate = 15/2/1]; m.p. 155.5–156.4 °C (petroleum ether/ethyl ether) (Lit^[14] m.p.: 160 °C); ¹H NMR (400 MHz, CDCl₃): δ = 8.14 (d, *J* = 2.4 Hz, 1H, ArH), 8.00 (t, *J* = 2.2 Hz, 1H, ArH), 7.60 (dd, *J*₁ = 8.8 Hz, *J*₂ = 2.4 Hz, 1H, ArH), 7.51–7.41 (m, 3H, ArH), 6.64 (s, 1 H, =CH); ¹³C NMR (100 MHz, CDCl₃): δ = 177.1, 159.7, 154.3, 133.83, 133.77, 131.1, 127.5, 127.1, 125.1, 125.0, 124.9, 119.6, 107.0; MS (70 eV, EI) *m/z* (%): 264 (M⁺ (³⁷Cl), 37.29), 262 (M⁺ (³⁵Cl), 100); IR (neat): ν = 1649, 1616, 1603, 1567, 1434, 1410, 1380, 1332, 1258, 1042 cm⁻¹

(3) Preparation of 2-(4-methoxyphenyl)-4H-chromen-4-one (**3c**)



Typical Procedure D: To a flame-dried Schlenk tube were added Fe(NO₃)₃·9H₂O (10.4 mg, 0.025 mmol) and NaCl (1.5 mg, 0.025 mmol) under argon. The Schlenk tube was then degassed to remove the argon inside completely, and refilled with O₂ by a balloon of O₂ for three times. Then, a solution of TEMPO (3.9 mg, 0.025 mmol) in dry MeCN (1.5 mL) and a solution of **1c** (127.5 mg, 0.5 mmol) in dry MeCN (1 mL) were added sequentially. Then the mixture was stirred at 25 °C with a balloon of O₂ for 3.75 h. After the disappearance of **1c** by TLC, the O₂ balloon was removed and DMAP (12.3 mg, 0.1 mmol) was added into the reaction mixture. Then the mixture was stirred at 25 °C for 5.5 h as monitored by TLC. After concentration under reduced pressure, the residue was transferred with DCM, filtered through a short column on silica gel (3 cm) (eluent: ethyl ether) until no product can be observed by TLC, and concentrated under reduced pressure [the yield of **3c** was determined by ¹H NMR analysis of the crude products using dibromomethane as the internal standard (78%)]. The residue was purified by column chromatography on silica gel to afford **3c**^[8] (102.5 mg, 81%) as a pale yellow solid (eluent: petroleum ether/ethyl acetate = 10/1 (210 mL) to petroleum ether/ethyl acetate = 5/1); m.p. 155.0–156.1 °C (petroleum ether/DCM) (Lit^[7e] m.p.: 154–156 °C); ¹H NMR (400 MHz, CDCl₃): δ = 8.23 (dd, *J*₁ = 8.2 Hz, *J*₂ = 1.4 Hz, 1H, ArH), 7.89 (d, *J* = 9.2 Hz, 2H, ArH), 7.73–7.65 (m, 1H, ArH), 7.56 (d, *J* = 8.0 Hz, 1 Hz, ArH), 7.45–7.37 (m, 1H, ArH), 7.03 (d, *J* = 9.2 Hz, 2H, ArH), 6.76 (s, 1H, =CH), 3.90 (s, 3H, OCH₃); ¹³C NMR (100 MHz, CDCl₃): δ = 178.4, 163.4, 162.4, 156.1, 133.5, 128.0, 125.6, 125.0, 123.9, 123.8, 117.9, 114.4, 106.1, 55.5; MS

(70 eV, EI) *m/z* (%): 252 (M⁺, 100); IR (neat): ν = 1644, 1604, 1570, 1512, 1464, 1426, 1376, 1314, 1264, 1251, 1192, 1122, 1023 cm⁻¹.

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